

Kinetics of the chemically activated HSO_5 radical under atmospheric conditions – a master-equation study

Núria González-García and Matthias Olzmann*

Received 20th April 2010, Accepted 14th July 2010

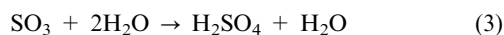
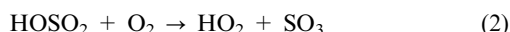
DOI: 10.1039/c0cp00284d

A detailed theoretical analysis of the $\text{HOSO}_2 + \text{O}_2$ reaction is performed, paying special attention to the kinetics of the intermediate HSO_5 radical. The possible formation of the monohydrated adduct, $\text{HSO}_5\cdot\text{H}_2\text{O}$, in the presence of water vapor is examined. For the binding energy of the most stable isomer at $T = 0\text{ K}$, a value of $D_0(\text{HOSO}_4\text{--H}_2\text{O}) = 51.7\text{ kJ mol}^{-1}$ was obtained at the CCSD(T)/CBS level of theory; other energies were adopted from a recently published high-level quantum chemical study of our laboratory. Molecular geometries and vibrational frequencies of the reactants, intermediates, and products were obtained from B3LYP/cc-pVTZ calculations. Rate coefficients and lifetimes of the HSO_5 intermediate were calculated by solving a master equation with specific rate coefficients from statistical rate theory. The master equation is extended by a bimolecular sink term, which accounts for the $\text{HSO}_5 + \text{H}_2\text{O}$ reaction. The relation between thermal and chemical activation in this reaction system is examined. The results show that the lifetime of the HSO_5 intermediate under atmospheric conditions is too short for a bimolecular reaction with water to become important. Relative yields of $\text{HSO}_5\cdot\text{H}_2\text{O}$ well below 1% were obtained, ruling out this reaction pathway as a relevant source for aerosols.

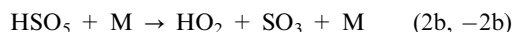
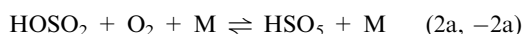
1. Introduction

The formation of aerosol particles in the atmosphere has received increasing attention in recent years. Aerosols have a direct impact on climate due to their optical properties but also act indirectly *e.g.*, as cloud condensation nuclei.^{1–3} Sulfate aerosols, which are formed at least in part from homogeneous nucleation of binary sulfuric acid/water ($\text{H}_2\text{SO}_4/\text{H}_2\text{O}$) mixtures, are known to be one of the major atmospheric aerosol components.^{1,2}

In the atmosphere H_2SO_4 is mainly produced by oxidation of SO_2 , which is known to essentially proceed along the following reaction sequence:^{1,2,4,5}

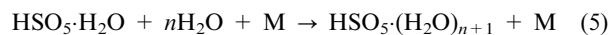


Here, eqn (2) is written as an overall reaction, but it is likely to proceed in a two-step addition-elimination mechanism:^{4,6,7}



Since reaction (2a) is exothermic by $\Delta_R H_{298\text{ K}}(2a) = -72.9 \pm 3.0\text{ kJ mol}^{-1}$ (for $\text{HSO}_5 = \text{HOSO}_4$, see below),⁷ the intermediate HSO_5 is chemically activated with a corresponding rovibrational excitation.

Recent laboratory investigations^{8–10} and theoretical studies^{11,12} provided indications that other sulfur-containing species formed along the SO_2 oxidation route could also contribute to particle formation, a hypothesis, which was already suggested by Friend *et al.*¹³ in 1980. These authors proposed a mechanism, which involved the formation and recombination of hydrated forms of HOSO_2 and HSO_5 radicals. This idea was resumed in a work by Berndt *et al.*⁹ to explain differences observed in experiments, where particle formation was induced in a flow reactor by direct sulfuric acid injection on the one hand and by *in situ* oxidation of SO_2 on the other hand.⁸ Very recently, Kurtén *et al.*¹⁴ proposed possible new pathways for atmospheric aerosol formation, assuming that the HSO_5 intermediate could be stabilized by reaction with water. The resulting adduct was thought to live long enough for further addition of water to form larger clusters:



These new pathways were investigated on the basis of extensive quantum chemical calculations and a global kinetic model.¹⁴ The authors concluded that the hydration product $\text{HSO}_5\cdot\text{H}_2\text{O}$ would indeed be sufficiently stable, but it remained an open question whether the chemically activated HSO_5 radical under atmospheric conditions has a unimolecular lifetime long enough to be directly hydrated in reaction (4). The previously assigned importance of these reaction pathways was questioned in a very recent publication¹⁵ in view of new experimental results on the basis of improved particle detection techniques. However, a conclusive picture about the relative contribution of these channels has not yet been established.

Institut für Physikalische Chemie, Karlsruher Institut für Technologie (KIT), Kaiserstr. 12, 76131 Karlsruhe, Germany.
E-mail: matthias.olzmann@kit.edu; Fax: +49-721-6086524

The rate coefficient of reaction (2) has been experimentally determined,^{16–19} and a good agreement between the different experiments at $T = 298$ K was obtained with values in the range $k_2 = (3.5\text{--}4.5) \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$. Gleason and Howard¹⁹ also studied the temperature dependence of k_2 between 297 and 423 K and obtained: $k_2(T) = (1.34 \pm 0.25) \times 10^{-12} \exp[(-2.7 \pm 0.6) \text{ kJ mol}^{-1}/RT] \text{ cm}^3 \text{ s}^{-1}$. Such a weak temperature dependence is an indication for an addition-elimination mechanism with a nearly barrierless addition step, eqn (2a).

Despite the large number of experimental and modeling studies, it is still not clear whether the HSO_5 intermediate can act as a precursor in aerosol formation. The key point is to characterize the lifetimes of this chemically activated intermediate with respect to unimolecular decomposition and bimolecular reaction with water under atmospheric conditions. If its decomposition into products, *via* reaction (2b), is too fast compared to hydration in reaction (4), the hypotheses based on the ability of HSO_5 to react with water molecules will need to be re-examined and other species/reaction routes for aerosol formation would have to be taken into account.

Our main objective in the present work is to calculate the lifetime of the HSO_5 intermediate under atmospheric conditions to clarify whether it can be considered a significant precursor for aerosol formation. A good energetic description of the reactions involved in the SO_2 atmospheric oxidation is a first essential point for a reliable prediction of the kinetic quantities. High-level quantum chemical calculations have been performed in our laboratory to determine the energies of both HOSO_2 and HSO_5 . On the basis of these recently published data,^{7,20} we will calculate in the present work the average lifetime of the HSO_5 intermediate and the relative branching fractions of its different reaction channels for varying conditions of temperature, pressure, and water concentration. One-dimensional master equations, including a sink term for the hydration reaction (4), have been set up and solved for both thermal and chemical activation conditions. Specific rate coefficients were calculated from statistical rate theory, and a step-ladder model obeying detailed balancing was employed for the transition probabilities. The overall rate coefficients obtained are analyzed and discussed with respect to atmospheric implications.

2. Methodology

2.1 Quantum chemical calculations

Calculated thermochemical parameters characterizing the unimolecular decomposition reactions of the HSO_5 radical in eqn (2a) and (2b) were published in a very recent work.⁷ It has been shown that it is necessary to apply both high-level coupled-cluster methods and large basis sets to obtain reliable results for the dissociation/association steps. The values obtained for the enthalpies of reactions of reaction (2a) and (2b) at $T = 0$ K (HOSO_4 isomer, see below) are $\Delta_R H_0^\circ \text{K}(2a) = -71.4 \pm 2.0 \text{ kJ mol}^{-1}$ and $\Delta_R H_0^\circ \text{K}(2b) = 62.9 \pm 5.0 \text{ kJ mol}^{-1}$.

In the present work, geometries and harmonic wavenumbers of the stationary points along the energy profiles of reactions (2) and (4) were obtained from density functional theory²¹

employing the Becke-3-Lee-Yang-Parr functional (B3LYP)^{22,23} with Dunning's correlation consistent basis set cc-pVTZ.²⁴ Wavenumbers were scaled by 0.9682.²⁵ The energies were obtained from single-point coupled cluster calculations^{26,27} with single and double excitations including triples corrections (CCSD(T))²⁸ with the cc-pVDZ or cc-pVTZ basis set.²⁴ The CCSD(T) energies at the complete basis set (CBS) limit were calculated by employing the extrapolation scheme developed by Helgaker *et al.*^{29,30} based on cc-pVDZ and cc-pVTZ basis sets.²⁴ All calculations were performed with the Gaussian03 program suite.³¹

In Fig. 1 the energy profile of the overall reaction (2) is displayed. We found that there are two isomers of HSO_5 . Obviously, HOSO_4 does not decompose directly to $\text{SO}_3 + \text{HO}_2$ but first isomerizes to $\text{SO}_3\text{-HO}_2$. The two isomers are connected by a transition state with a low energy barrier. We also found a transition state for a direct H abstraction from HOSO_2 by O_2 ($\text{TS}_{\text{direct}}$, not shown in Fig. 1) with an energy of $136.6 \text{ kJ mol}^{-1}$ relative to $\text{HOSO}_2 + \text{O}_2$ at $T = 0$ K, which is too high for a direct abstraction reaction to play any significant role under atmospheric conditions. The molecular geometries of these species are displayed in Fig. 2, and selected parameters are collected in Tables 1 and 2. Note that we could not find transition states for reaction (2a) and (2b).

To characterize the hydration reaction, eqn (4), we calculated the binding energy of the most stable $\text{HSO}_5\text{-H}_2\text{O}$ adduct at the CCSD(T)/CBS level of theory; we obtained a value of $D_0(\text{HOSO}_4\text{-H}_2\text{O}) = 51.7 \text{ kJ mol}^{-1}$ for $T = 0$ K. The optimized geometry is shown in Fig. 2, and selected bond lengths are given in Table 1. The adduct represents a structure with one strong hydrogen bond ($d = 1.57 \text{ \AA}$) between the oxygen of the H_2O moiety and the hydrogen of the HOSO_4 moiety. Hydrogen bonds involving the H atoms of the water moiety are also formed but exhibit longer bond distances of $2.29 \text{ \AA}$ and $3.04 \text{ \AA}$. The binding energy we obtained is in reasonable agreement with the results of Kurtén *et al.*¹⁴ Adopting their values for the potential energy difference (ΔE_{elec})¹⁴ and employing the zero-point energies from the present work, we obtain $D_0(\text{HOSO}_4\text{-H}_2\text{O}) = 62.9 \text{ kJ mol}^{-1}$

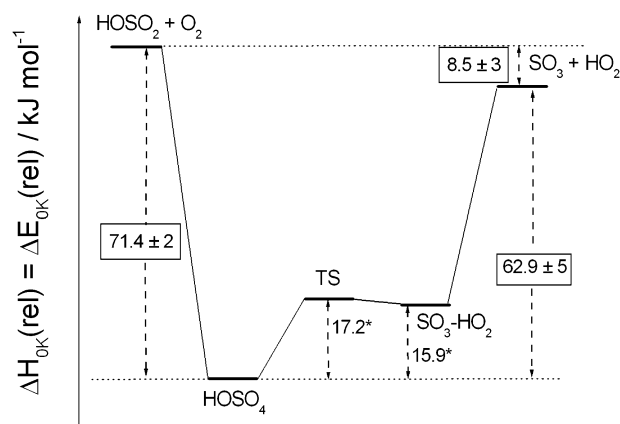


Fig. 1 Energy diagram of HSO_5 decomposition including zero-point energies (units kJ mol^{-1}). The energies marked with star (*) have been calculated at the CCSD(T)/CBS level of theory in this work, the values in boxes were adopted from ref. 7 and 20.

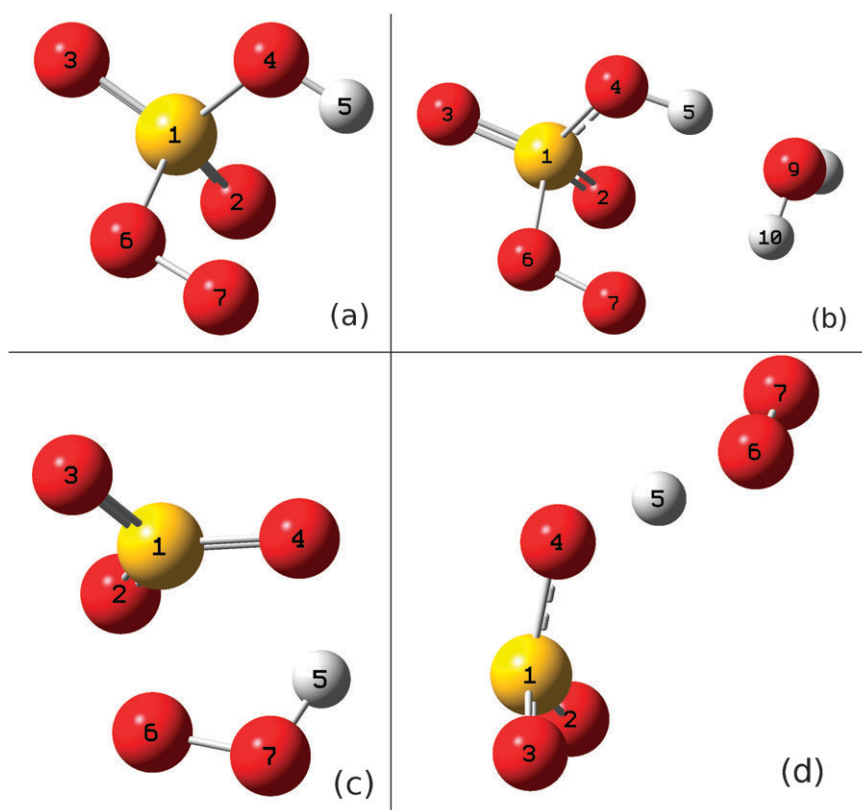


Fig. 2 Geometries of the HOSO₄ radical (a), HOSO₄·H₂O adduct (b), SO₃·HO₂ (c) and TS_{direct} (d) optimized at the B3LYP/cc-pVTZ level of theory; for bond distances see Table 1.

Table 1 Equilibrium bond lengths (in Å) of different species along the SO₂ oxidation route, for numbering of atoms see Fig. 2

Parameter	HOSO ₄	HOSO ₄ ·H ₂ O	TS _{direct}	SO ₃ ·HO ₂
S–O ₂	1.43	1.44	1.45	1.44
S–O ₃	1.43	1.43	1.45	1.44
S–O ₄	1.59	1.56	1.52	1.46
O ₄ –H ₅	0.97	1.02	1.21	1.71
S–O ₆	1.76	1.78	—	2.21
O ₆ –O ₇	1.31	1.31	1.26	1.31
H ₅ –O ₉	—	1.57	—	—
O ₇ –H ₁₀	—	2.29	—	—
O ₂ –O ₈	—	3.04	—	—

(G3B3)¹⁴ and $D_0(\text{HOSO}_4\text{--H}_2\text{O}) = 50.7 \text{ kJ mol}^{-1}$ (CCSD(T)/aug-cc-pV(T+d)Z//B3LYP/6-311++G(3df,3pd)),¹⁴ respectively.

We note that HOSO₄·H₂O is more stable than HOSO₂·H₂O previously described by Aaltonen and Francisco.³² The binding energy of the latter adduct was calculated at the G3 level of theory to be $D_0(\text{HOSO}_2\text{--H}_2\text{O}) = 42.7 \text{ kJ mol}^{-1}$.³² In HOSO₂·H₂O the third hydrogen bond cannot be formed, and therefore the binding energy is expected to be somewhat smaller than in HOSO₄·H₂O.

2.2 Statistical rate theory calculations

Master equation. As already mentioned above, the HSO₅ radical formed in the exothermic reaction (2a) is chemically activated. Rate coefficients for unimolecular reactions of such species are most rigorously calculated by solving an appropriate master equation (see *e.g.*, ref. 33,34). The master

equation, that describes the energetic distribution of HSO₅ as an intermediate in reactions (2a), (2b), and (4) reads as:

$$\frac{dn(E)}{dt} = R_{2a}f(E) - \omega n(E) + \omega \sum_{\varepsilon} P(E, \varepsilon)n(\varepsilon) - (k_{-2a}(E) + k_{2b}(E) + k_4[\text{H}_2\text{O}])n(E) \quad (6)$$

where $n(E)$ is the concentration of HSO₅ species having internal energy E . The quantity R_{2a} is the overall rate of HSO₅ formation in reaction (2a), and $f(E)$ denotes the normalized nascent distribution of the species formed. The second and third terms describe the collisional depopulation and population, respectively, of the considered energy level E with the collision frequency ω and the transition probabilities $P(E, \varepsilon)$ for transitions $\varepsilon \rightarrow E$. The next two terms represent the unimolecular decomposition reaction (–2a) and (2b) with the specific rate coefficients $k_{-2a}(E)$ and $k_{2b}(E)$, respectively, while the bimolecular reaction (4) is accounted for by its pseudo-first order rate coefficient $k_4[\text{H}_2\text{O}]$.^{35–38} In this way, the competition between the unimolecular decomposition channels of HSO₅ and its consecutive bimolecular reaction with H₂O is consistently incorporated in the master equation. The general kinetics of such competing reactions in chemical activation systems is examined in detail in ref. 36.

Eqn (6) can be cast into matrix form,^{33,34} and by assuming steady-state conditions, $dn(E)/dt = 0$, one obtains

$$R_{2a}\mathbf{F} = [\omega(\mathbf{I} - \mathbf{P}) + \mathbf{K}_{-2a} + \mathbf{K}_{2b} + k_4[\text{H}_2\text{O}]\mathbf{I}]\mathbf{N}^s \equiv \mathbf{J}\mathbf{N}^s \quad (7)$$

where the vector/matrix symbols correspond to the symbols in eqn (6), and $\mathbf{I}$ represents the unit matrix. The normalized vector of the steady-state population, $\tilde{\mathbf{N}}^s$, formally follows from

$$\tilde{\mathbf{N}}^s = \frac{\mathbf{J}^{-1}\mathbf{F}}{\sum_i (\mathbf{J}^{-1}\mathbf{F})_i} \quad (8)$$

where $(\mathbf{X})_i$ denotes the i th element of a vector $\mathbf{X}$. From the steady-state population $\tilde{\mathbf{N}}^s$, the observable rate coefficient of a chemically activated (superscript: ca) unimolecular reaction channel r is obtained by averaging the specific rate coefficient $k_r(E)$ over $n^s(E)$; in vector/matrix notation one has:

$$k_r^{\text{ca}}(T, P) = \sum_i (\mathbf{K}_r \tilde{\mathbf{N}}^s)_i \quad (9)$$

The relative yield of a given reaction channel is then obtained from the corresponding ratio of rate coefficients. In particular, the relative yield of reaction (4) follows from

$$\phi_4(T, P) = \frac{k_4[\text{H}_2\text{O}]}{k_{-2a}^{\text{ca}}(T, P) + k_{2b}^{\text{ca}}(T, P) + k_4(\text{H}_2\text{O})} \quad (10)$$

The overall rate coefficient for the chemically activated decomposition of HSO_5 can be written as

$$k^{\text{ca}}(T, P) = k_{-2a}^{\text{ca}}(T, P) + k_{2b}^{\text{ca}}(T, P) \quad (11)$$

It is important to note that also the rate coefficient of the thermal (superscript: th) unimolecular decomposition of HSO_5 can be obtained from eqn (7) as the lowest eigenvalue λ_1 of the matrix $\mathbf{J}$ for $[\text{H}_2\text{O}] = 0$ (due to our definition of $\mathbf{J}$, the eigenvalues are positive):³³

$$k^{\text{th}}(T, P) = k_{-2a}^{\text{th}}(T, P) + k_{2b}^{\text{th}}(T, P) = \lambda_1 \quad (12)$$

Alternatively, the rate coefficient $k_i^{\text{th}}(T, P)$ can also be calculated by an averaging procedure analogous to eqn (9) but with $\tilde{\mathbf{N}}^s = \tilde{\mathbf{N}}_{\text{th}}^s$ being the normalized eigenvector associated with the lowest eigenvalue λ_1 .³³

The stepladder model used for $P(E, \varepsilon)$ (see below) causes the matrix $\mathbf{J}$ to be tridiagonal, and $\tilde{\mathbf{N}}^s$ was obtained by solving eqn (7) for $\mathbf{N}^s$ with standard procedures (routines *banbks* and *bandec* from ref. 39) and subsequent normalization. The eigenvalues λ_i and eigenvectors $\tilde{\mathbf{N}}_{\text{th}}^s$ of $\mathbf{J}$ were calculated by using the EISPACK routine *tql2*⁴⁰ (see also ref. 39) after symmetrization of $\mathbf{J}$.³³ The master equation was set up with a bin size of 10 cm^{-1} , and more technical details were already given in ref. 36. The different quantities entering eqn (6) and (7) will be briefly discussed in the following.

Specific rate coefficients. In our quantum chemical calculations, we did not find any transition states for reactions (−2a) and (2b). Accordingly, the specific rate coefficients $k_{-2a}(E)$ and $k_{2b}(E)$ were calculated from the simplified statistical adiabatic channel model (SACM).^{41,42}

$$k_r(E) = \frac{W_r(E - E_{0(r)})}{h\rho(E)} \quad (13)$$

with $W_r(E - E_{0(r)})$ being the number of open reaction channels of reaction r , and $\rho(E)$ denoting the density of states of the reactant/intermediate, $E_{0(r)}$ is the threshold energy of reaction r at $T = 0 \text{ K}$, and h stands for Planck's constant.

In calculating $\rho(E)$, we assumed that the interconversion of the two isomers, $\text{HOSO}_4 \rightleftharpoons \text{SO}_3 \cdot \text{HO}_2$, is much faster than reactions (−2a) and (2b) due to the much lower energy barrier (see Fig. 1). Therefore, reactions (−2a) and (2b) occur from an equilibrated mixture of the two isomers, and the densities of states have to be combined:⁴³

$$\rho(E) = \rho_{\text{HOSO}_4}(E) + \rho_{\text{SO}_3 \cdot \text{HO}_2}(E) \quad (14)$$

To calculate the numbers of open reaction channels $W_r(\varepsilon)$, one has to specify the interpolation parameter α/β of the statistical adiabatic channel model.^{41,42,44} This was accomplished for reaction (−2a) by fitting the rate coefficient $k_{2a}^{\infty}(T)$ calculated by the *canonical* version of SACM^{41,44} to the experimental value of $k_2(T)$ using the ratio α/β as a parameter. Since α/β is the same for reactions (2a) and (−2a) due to detailed balancing, the value obtained was then used to calculate $W_{(-2a)}(\varepsilon)$ with the *microcanonical* version of SACM.⁴² The relation $k_{2a}^{\infty}(T) \sim k_2(T)$ with $k_2(T) \neq f(P)$ holds, because $k_{-2a}(E) \ll k_{2b}(E)$ in the relevant energy range (see below). This situation is well known for exothermic association-elimination reactions.⁴⁷

For an optimum value of $\alpha/\beta = 0.24$, we obtained a rate coefficient $k_{2a}^{\infty}(T = 298 \text{ K}) = 4.1 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ and an activation energy of $E_{a(2a)} = 3.6 \text{ kJ/mol}$. These values agree well with the experimental results for $k_2(T)$ located in the range $(3.5 - 4.5) \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ ^{16–19} and with the experimental energy of activation $E_{a(2)} = (2.7 \pm 0.6) \text{ kJ mol}^{-1}$ obtained by Gleason and Howard,¹⁹ respectively. In contrast to reaction (2a), no experimental kinetic data related to reaction (2b) are available, and the SACM calculations for $k_{2b}(E)$ had to be performed with the recommended “standard value” of $\alpha/\beta = 0.5$,^{44,45} which corresponds to a high-pressure limit of $k_{2a}^{\infty}(T = 298 \text{ K}) = 1.3 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$. In ref. 44, it was shown that values of α/β between 0.31 and 0.61 could reproduce experimental high-pressure rate coefficients of 26 different bond fission/association reactions. Furthermore, in a comparison of results from the exact and simplified SACM with experimental rate coefficients, a value of $\alpha/\beta = 0.5$ was found to give a satisfactory agreement.⁴⁵ Even though we had to use a smaller value of $\alpha/\beta = 0.24$ to characterize reaction (2a, −2a), such a low value for reaction (2b, −2b) is unlikely, because the $\text{HO}_2 + \text{SO}_3$ reaction was recently shown^{12,46} to lead to an adduct, that exhibits the typical features of a weakly bound complex. This normally gives rise to loose critical configurations/transition states and hence to rather large values of α/β . Completely analogous arguments apply to reaction (4). The influence of a variation of α/β on our results is discussed in section 3.

Densities of states $\rho(\varepsilon)$ and numbers of open reaction channels $W_r(\varepsilon)$ were determined by direct counting procedures^{42,48,49} in the rigid rotor/harmonic oscillator approximation for a total angular momentum quantum number $J = 49$, corresponding to the average rotational excitation of HOSO_4 at $T = 298 \text{ K}$. The influence of varying J on our results is vanishingly small and will be discussed in section 3. Threshold energies $E_{0(r)}(J = 49)$ were determined on the basis of Morse functions for the interfragment potentials and the quasi-diatomic approximation for the centrifugal energy as described in

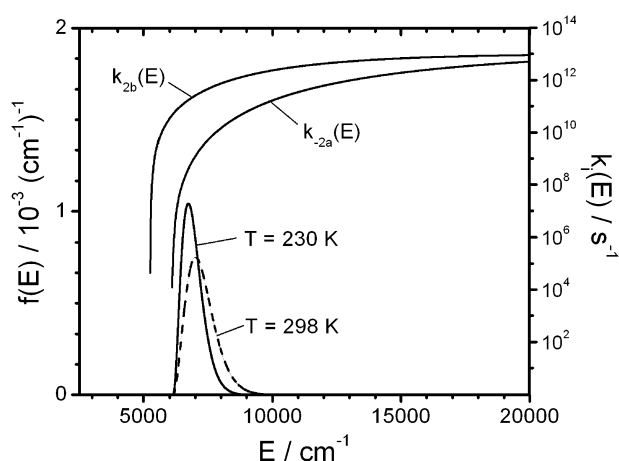


Fig. 3 Nascent population of HSO_5 at $T = 230$ K (solid curve) and $T = 298$ K (dashed curve) and specific rate coefficients of the unimolecular reactions (–2a) and (2b).

ref. 42. Adopting $\Delta_R H_{0\text{K}}(-2a) = 71.4 \text{ kJ mol}^{-1}$ and $\Delta_R H_{0\text{K}}(2b) = 62.9 \text{ kJ mol}^{-1}$,⁷ we obtained $E_{0(-2a)} = 73.1 \text{ kJ/mol}$ and $E_{0(2b)} = 63.1 \text{ kJ/mol}$. In Fig. 3 the specific rate coefficients for back dissociation, $k_{-2a}(E)$, and forward reaction, $k_{2b}(E)$, of HSO_5 are shown. The frequencies and rotational constants used in our calculations are listed in Table 2.

Nascent distribution. The initial population of the HSO_5 intermediate formed in reaction (2a) can be calculated from the number of open reaction channels of reaction (2a), $W_{2a}(\varepsilon) = W_{-2a}(\varepsilon)$:^{33,34,50}

$$f(E) = \frac{W_{-2a}(E - E_{0(-2a)}) \exp[-(E - E_{0(-2a)})/k_B T]}{\int_0^\infty W_{-2a}(\varepsilon) \exp(-\varepsilon/k_B T) d\varepsilon} \quad (15)$$

where k_B denotes Boltzmann's constant. In Fig. 3 this distribution is displayed for two temperatures enclosing the tropospherically relevant temperature range, viz. 230 K and 298 K.

Collisional energy transfer. The rate coefficient of collisional energy transfer is expressed in eqn (6) in the form $\omega P(E, \varepsilon)$. For the calculation of the collision frequency of HSO_5 in air, $\omega = Z_{\text{LJ}} P$, with the Lennard-Jones collision number Z_{LJ} and the pressure P , we approximated the Lennard-Jones parameters of HSO_5 , using the relations given by Stiel and Todos⁵¹ and critical parameters of H_2SO_4 from ref. 52; the data for N_2 and O_2 were taken from ref. 53. We obtained $Z_{\text{LJ}} = 9.7 \times 10^6 \text{ mbar}^{-1} \text{ s}^{-1} \sim 1 \times 10^7 \text{ mbar}^{-1} \text{ s}^{-1}$.

For the transition probabilities, $P(E, \varepsilon)$, the stepladder model obeying detailed balancing was used.^{33,34} The step size, ΔE_{SL} , which represents the average amount of energy transferred in down collisions, can be related to the average energy transferred in both up and down collisions, $\langle \Delta E \rangle$, by^{54,56}

$$\langle \Delta E \rangle = -\Delta E_{\text{SL}} \tanh\left(\frac{\Delta E_{\text{SL}}}{2F_E k_B T}\right) \quad (16)$$

where F_E characterizes the energy dependence of the density of states.⁵⁷ We tested three different values of ΔE_{SL} , viz., 50 cm^{-1} , 100 cm^{-1} , and 200 cm^{-1} . These cover a reasonable range in view of the moderate excitation of HSO_5 in the order of 7000 cm^{-1} .⁵⁸ For the HSO_5 intermediate at temperatures near 300 K ($F_E \sim 1$), these values correspond to average energies transferred per collision of $\langle \Delta E \rangle = -6.0 \text{ cm}^{-1}$, -24 cm^{-1} , and -90 cm^{-1} , respectively.

The rate of collisional stabilization can be conveniently approximated by an effective collision frequency $\gamma_c \omega$ with a

Table 2 Rotational constants (A , B , C) and harmonic wavenumbers (ω_i) of stationary points calculated at the B3LYP/cc-pVTZ level of theory. The wavenumbers have been scaled by 0.9682²⁵

Species	A , B , C/cm^{-1}	ω_i/cm^{-1}
HOSO_2	0.30178	269, 407, 415, 514, 735,
	0.29487	1077, 1111, 1275, 3742
	0.15931	
O_2	1.44958	1629
	0.15966	92, 197, 303, 321, 351, 426,
	0.09247	469, 542, 617, 815, 1102,
HOSO_4	0.09071	1120, 1169, 1407, 3611
	0.17146	i884, 128, 256, 275, 457, 481,
	0.08754	576, 590, 727, 999, 1043,
TS (see Fig. 1)	0.08645	1201, 1236, 1359, 1824
	0.16724	90, 173, 206, 248, 398, 495,
	0.07890	504, 515, 719, 991, 1190,
$\text{SO}_3 \cdot \text{HO}_2$	0.07860	1259, 1345, 1499, 3001
	0.33937	462, 496, 1017, 1330, 1331
	0.33937	
SO_3	0.16968	
	20.68495	1123, 1387, 3488
	1.12318	
HO_2	1.06533	
	27.21636	1587, 3682, 3779
	14.47505	
H_2O	9.44939	
	0.10013	66, 100, 105, 182, 237, 246, 321, 338, 398,
	0.06100	452, 474, 509, 549, 613, 879, 1034, 1127,
$\text{HOSO}_4 \cdot \text{H}_2\text{O}$	0.04322	1163, 1314, 1414, 1589, 2702, 3621, 3724

collision efficiency $\gamma_c < 1$. Following ref. 55, we estimate this collision efficiency from the relation

$$\frac{\gamma_c}{1 - \gamma_c^2} = \frac{-\langle \Delta E \rangle s^*}{\langle E \rangle - E_{0(2b)}} \quad (17)$$

where $s^* = 3.4$ characterizes the energy dependence of $k_{(2b)}(E)$,⁵⁵ and $\langle E \rangle = 7000 \text{ cm}^{-1}$ represents the average energy of $f(E)$. We obtain $\gamma_c \sim 0.01, 0.05$, and 0.17 for $\Delta E_{\text{SL}} = 50 \text{ cm}^{-1}$, 100 cm^{-1} , and 200 cm^{-1} , respectively. It will be shown below, that the atmospheric fate of HSO_5 does not change if ΔE_{SL} is varied within this range. The exemplary calculations in the following sections are performed with $\Delta E_{\text{SL}} = 100 \text{ cm}^{-1}$.

3. Results and discussion

It is instructive to solve the master equation for chemical and thermal activation and compare the results to understand the differences between the two regimes. At first, in sections 3.1 and 3.2, we will study the kinetics of HSO_5 in the absence of water, that is, we solve eqn (7) for $[\text{H}_2\text{O}] = 0$. Then, in section 3.3, the effects of finite water concentrations, $[\text{H}_2\text{O}] \neq 0$, will be examined.

The macroscopic rate coefficients $k_A(T, P)$ can be obtained in general by averaging specific rate coefficients $k_A(E)$ over the normalized steady-state distribution of the reacting species, $\tilde{n}^s(E)$ (cf. eqn (9)). In the case of chemically activated decomposition of HSO_5 produced in reaction (2a), the distribution function $\tilde{n}^s(E; T, P)$ is obtained from eqn (8), and in the case of thermal decomposition, the distribution function $\tilde{n}_{\text{th}}^s(E; T, P)$ is obtained as the normalized eigenvector associated with the lowest eigenvalue of the matrix **J**. These distribution functions are shown in Fig. 4 for $T = 298 \text{ K}$ and atmospheric pressure; the pressure-dependent rate coefficients are displayed in Fig. 5. The striking differences will be discussed in the following sections.

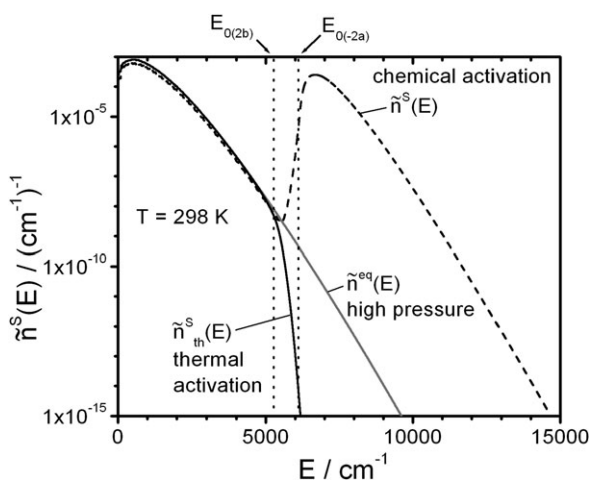


Fig. 4 Comparison of the normalized distribution functions of HSO_5 for chemical (dashed curve) and thermal (solid curve) activation at $T = 298 \text{ K}$, $P = 1 \text{ bar}$ with $\Delta E_{\text{SL}} = 100 \text{ cm}^{-1}$; also shown is the equilibrium distribution at high pressures (gray curve).

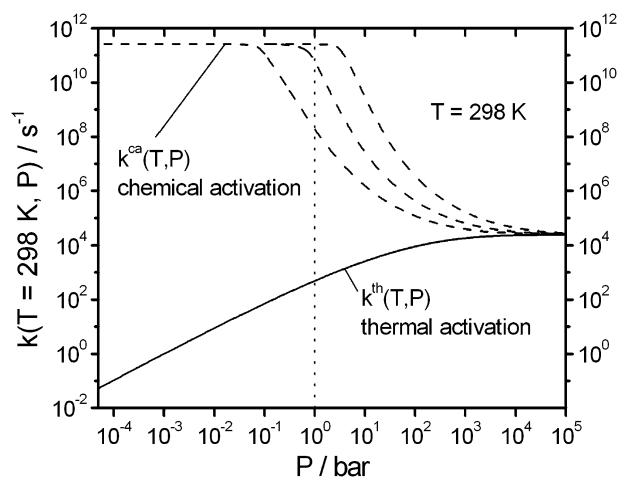


Fig. 5 Comparison of the rate coefficients of thermally (solid curve, $\Delta E_{\text{SL}} = 100 \text{ cm}^{-1}$) and chemically (dashed curves, $\Delta E_{\text{SL}} = 50, 100$, and 200 cm^{-1} , from top to bottom) activated HSO_5 decomposition ($T = 298 \text{ K}$).

3.1 Thermal decomposition of HSO_5

For the case of thermal activation, the familiar falloff curve of a unimolecular reaction can be realized in Fig. 5. It is obvious that the thermal decomposition of HSO_5 at room temperature and atmospheric pressure is in the falloff range close to second order. For $P = 1 \text{ bar}$, we obtain rate coefficients of $k_{-2a}^{\text{th}} = 1.4 \times 10^{-7} \text{ s}^{-1}$ and $k_{2b}^{\text{th}} = 5.0 \times 10^2 \text{ s}^{-1}$. Obviously, reaction (2b) with the lower threshold energy strongly dominates over reaction (−2a) and completely governs the overall decomposition. This is essentially due to the much higher specific rate coefficients $k_{2b}(E) \gg k_{-2a}(E)$ as can be seen in Fig. 3. One should also note that the population under thermal conditions, $\tilde{n}_{\text{th}}^s(E)$, is strongly depleted for energies $E > E_{0(2b)}$ (see Fig. 4), which causes the reaction to be in the falloff range. From the thermal rate coefficients, the average thermal lifetime of HSO_5 can be calculated: $\tau_{\text{HSO}_5}^{\text{th}}(T, P) = (k_{-2a}^{\text{th}}(T, P) + k_{2b}^{\text{th}}(T, P))^{-1}$ and for $P = 1 \text{ bar}$ and $T = 298 \text{ K}$, we obtain $\tau_{\text{HSO}_5}^{\text{th}} = 2.0 \text{ ms}$.

3.2 Chemically activated decomposition of HSO_5

In a time-dependent picture of chemical activation, the average thermal lifetime of the adduct characterizes the time-scale on which the final or asymptotic steady state⁵⁹ is reached (see also the discussions in ref. 33,36,60–62). In contrast to the so called intermediate steady state,⁵⁹ which is characterized by the competition between decomposition and stabilization of the chemically activated adduct, the term final steady state implies that there is no more net stabilization; the stabilization reservoir is filled up, and time-independent energy distributions have been established. This allows the use of rate coefficients instead of eigenvalue-eigenvector expansions to describe the kinetic behavior.^{33,62}

In technical terms the intermediate steady state can be implemented by introducing a lower absorbing barrier into the master equation.³⁴ By omitting this absorbing barrier, that is by carefully observing the completeness of transition probabilities, the final steady-state solution for $\tilde{n}^s(E)$ is obtained from eqn (8). We assume that under atmospheric conditions

Table 3 Calculated lifetimes (unit: s) of chemically activated HSO₅

P/bar	$\Delta E_{\text{SL}}/\text{cm}^{-1}$	$T = 230 \text{ K}$	$T = 298 \text{ K}$
$P \rightarrow 0$		5.4×10^{-12}	3.8×10^{-12}
$P \rightarrow \infty$		8.9×10^{-2}	4.0×10^{-5}
1	50	5.9×10^{-12}	3.8×10^{-12}
1	100	4.5×10^{-8}	1.5×10^{-11}
1	200	2.4×10^{-5}	6.0×10^{-9}

the concentrations of the reactants of reaction (2a), HOSO₂ and O₂, do not drastically change on a timescale of 2 ms (the thermal lifetime of HSO₅, see above) and, therefore, use the final steady-state approximation for the kinetic description within this work.

The resulting pressure dependence of the rate coefficient for the decomposition of chemically activated HSO₅ formed in reaction (2a) at $T = 298 \text{ K}$ is displayed in Fig. 5. One can realize a pressure-independent low-pressure limit ($k_0^{\text{ca}} = 2.6 \times 10^{11} \text{ s}^{-1}$), where collisional deactivation is too slow to compete with the unimolecular decomposition of the excited adduct. Then, at pressures between 0.1 and 10 bar, depending on ΔE_{SL} , collisional deactivation starts to become effective, and the rate coefficient drops; finally at high pressures, collisional stabilization strongly dominates, and the rate coefficient approaches the thermal high-pressure limit. The drop from the low-pressure limit occurs for all values of ΔE_{SL} at effective collision numbers of $\gamma_c \omega \sim 2 \times 10^8 \text{ s}^{-1}$. We further note a rather strong dependence on ΔE_{SL} of the rate coefficient in the intermediate pressure range, which is also reflected in the lifetimes collected in Table 3.

For atmospheric conditions ($T = 298 \text{ K}$ and $P = 1 \text{ bar}$) with $\Delta E_{\text{SL}} = 100 \text{ cm}^{-1}$, we obtain values of $k_{-2a}^{\text{ca}} = 6.0 \times 10^8 \text{ s}^{-1}$ and $k_{2b}^{\text{ca}} = 6.6 \times 10^{10} \text{ s}^{-1}$, which are much higher than for thermal excitation. For the average lifetime of the chemically activated HSO₅ under these conditions, it follows: $\tau_{\text{HSO}_5}^{\text{ca}} = (k_{-2a}^{\text{ca}} + k_{2b}^{\text{ca}})^{-1} = 15 \text{ ps}$, eight orders of magnitude lower than under thermal conditions. The reason for this difference are the completely different molecular distributions, which are displayed in Fig. 4. Since HSO₅ is formed in reaction (2a) with an internal energy of about 7000 cm^{-1} (the average energy of $f(E)$, cf. Fig. 3), higher energies can remain populated. For $P = 1 \text{ bar}$, the bimodal distribution shown in Fig. 5 arises. At this pressure, which corresponds to an effective collision number of $\gamma_c \omega \sim 5 \times 10^8 \text{ s}^{-1}$, collisional stabilization leads to a population at energies below $E_{0(2b)}$, that closely resembles the population under thermal conditions (see Fig. 4). At energies above $E_{0(2b)}$, however, the incoming flux from reaction (2a) increases the population distinctly, which leads to a second maximum and upon averaging to the much higher rate coefficient.

For vanishing collisional stabilization that is, for $P \rightarrow 0$, one would have $\tilde{n}^{\text{s}}(E) \rightarrow f(E)$; the low-pressure limit of the steady-state distribution is identical to the nascent distribution. For $P \rightarrow \infty$, collisional stabilization is complete, and the distributions for thermal as well as chemical activation turn into the equilibrium distribution $\tilde{n}^{\text{eq}}(E)$ (see Fig. 4). This results, as already mentioned above, in identical high-pressure limiting rate coefficients for thermal and chemical activation, which is illustrated Fig. 5. For $T = 298 \text{ K}$, we obtain $k_{-2a}^{\infty} = 3.1 \times 10^1 \text{ s}^{-1}$ and $k_{2b}^{\infty} = 2.5 \times 10^4 \text{ s}^{-1}$.

Average lifetimes of chemically activated HSO₅ intermediates produced in reaction (2a) for different conditions are collected in Table 3. The temperatures were chosen so as to span the tropospheric relevant range between planetary boundary layer ($T = 298 \text{ K}$) and regions near the tropopause ($T = 230 \text{ K}$).^{1,2} One can realize that the lifetimes at atmospheric pressure depend on ΔE_{SL} that is, the efficiency of collisional energy transfer, because the rate coefficient for unimolecular decomposition is in the pressure-dependent range (cf. Fig. 5).

3.3 Influence of water vapor

The bimolecular reaction of HSO₅ with H₂O, reaction (4), represents an additional reaction pathway for HSO₅ besides the unimolecular decomposition channels (–2a) and (2b). As we have shown in ref. 36, the existence of such a competing bimolecular step can lead to a change of the steady-state regime. The crucial quantity here is the pseudo-first order rate coefficient $k_4[\text{H}_2\text{O}]$, which characterizes the outgoing flux of HSO₅ from the intermediate population $\tilde{n}^{\text{s}}(E)$. If reaction (4) is fast enough, the stabilized HSO₅ population is prevented from being built up, and the applicability of the intermediate steady-state approximation is not temporally limited. To quantify the effect of water, the assumption $[\text{H}_2\text{O}] = 0$ in eqn (7), which was implied in sections 3.1 and 3.2, is now abandoned.

The formation of HSO₅·H₂O in reaction (4) is a bimolecular association step, where the HSO₅ population is at least in part rovibrationally excited. Accordingly, a most rigorous treatment would require to employ energy-dependent rate coefficients $k_4(E)$. As little is known about the energy dependence of rate coefficients for bimolecular reactions of polyatomic species, we use the thermal capture rate coefficient, k_4^{∞} , as an upper limit. This, in turn, would give an upper limit for the yield of reaction (4).

The rate coefficient $k_4^{\infty}(T)$ is calculated from the canonical version of the simplified SACM.^{41,44} The necessary molecular data of the reactants and product of reaction (4) are collected in Table 2, and for the SACM interpolation parameter, we used again the recommended value of $\alpha/\beta = 0.5$.⁴⁴ For the binding energy of HOSO₄·H₂O at $T = 0 \text{ K}$, a value of 51.7 kJ mol^{-1} was obtained from quantum chemical calculations at the CCSD(T)/CBS level of theory as outlined in section 2.1. Since the temperature dependence $k_4^{\infty}(T)$ is weak, we set in all our calculations $k_4 = k_4^{\infty}(T = 298 \text{ K}) = 9.9 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1} \sim 1 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$. The resulting pseudo-first order rate coefficients $k_4[\text{H}_2\text{O}]$ for different water concentrations are collected in Table 4. The relative humidities (RH) were calculated on the basis of vapor pressures from standard tables.⁶³ The value RH = 22% is included, because it represents the most common value in the experiments of Berndt *et al.*^{8,14}

With these rate coefficients, relative yields of reaction (4) can be calculated from eqn (10), and the results are also contained in Table 4. The pressure dependence is shown in Fig. 6. One can realize a pressure-independent low-pressure limit, which is proportional to the water concentration, a pressure-independent yield of unity for high pressures, and a yield increasing with

Table 4 Relative yield of $\text{HOSO}_4 \cdot \text{H}_2\text{O}$ as a function of temperature, pressure, and relative humidity (RH)

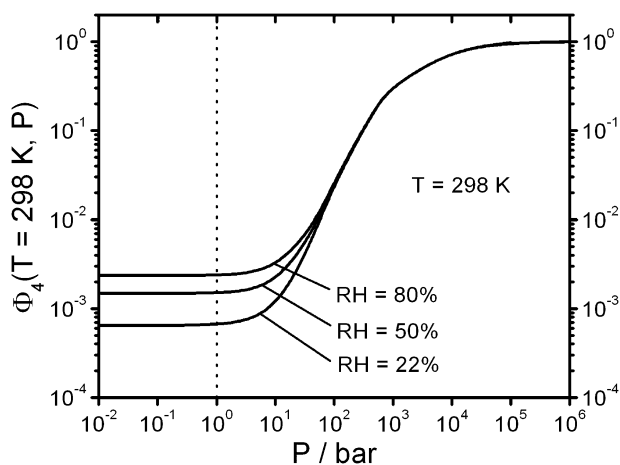
RH (%)	$[\text{H}_2\text{O}]/\text{cm}^{-3}$	$k_4[\text{H}_2\text{O}]/\text{s}^{-1}$	ϕ_4	
			$P = 1 \text{ bar}$	$P = 100 \text{ bar}$
$T = 298 \text{ K}$				
22	1.7×10^{17}	1.7×10^8	6.5×10^{-4}	2.5×10^{-2}
50	3.9×10^{17}	3.9×10^8	1.5×10^{-3}	2.6×10^{-2}
80	6.2×10^{17}	6.2×10^8	2.4×10^{-3}	2.7×10^{-2}
$T = 230 \text{ K}$				
22	6.2×10^{14}	6.2×10^5	2.2×10^{-6}	7.7×10^{-2}
50	1.4×10^{15}	1.4×10^6	6.3×10^{-6}	7.7×10^{-2}
80	2.3×10^{15}	2.3×10^6	1.1×10^{-5}	7.7×10^{-2}

pressure from the low-pressure limit to unity in the intermediate range. We note in passing that at pressures distinctly above 100 bar, the simple picture of binary collisions of course breaks down, and quantitative conclusions cannot longer be drawn from our model. In this pressure range our master equation merely reflects the correct mathematical behavior under the assumption of binary collisions.

In the following we will discuss the s-shaped pressure dependence and atmospheric implications. This is most easily done in terms of eqn (10) and the pressure dependence of the chemical activation rate coefficient $k^{\text{ca}}(T, P) \sim k_{2b}^{\text{ca}}(T, P) \gg k_{-2a}^{\text{ca}}(T, P)$ shown in Fig. 5. It is obvious that the pressure dependence of ϕ_4 reflects the pressure dependence of k^{ca} . One should note that under tropospheric conditions, the pseudo-first order rate coefficients $k_4[\text{H}_2\text{O}]$ are in the order of 10^8 s^{-1} .

In the low-pressure limit we have $k^{\text{ca}}(T, P \rightarrow 0) \gg k_4[\text{H}_2\text{O}]$, and it follows from eqn (10): $\phi_4(T, P \rightarrow 0) = k_4[\text{H}_2\text{O}]/k^{\text{ca}}(T, P \rightarrow 0)$, a pressure-independent yield, which is proportional to $[\text{H}_2\text{O}]$. In the high-pressure limit with $k^{\text{ca}}(T, P \rightarrow \infty) \ll k_4[\text{H}_2\text{O}]$, one obtains $\phi_4(T, P \rightarrow \infty) = 1$. The transition range between these two limits is pressure-dependent with a decreasing influence of water concentration with increasing pressure. This is quantitatively displayed in Fig. 6.

From Table 4 and Fig. 6, one can also realize that the yield of reaction (4) at atmospheric pressure depends on the concentration of water but stays well below 1% under all conditions.

**Fig. 6** Yield of reaction (4) at $T = 298 \text{ K}$ as a function of pressure at relative humidities of 22%, 50%, and 80% ($\Delta E_{\text{SL}} = 100 \text{ cm}^{-1}$).

The picture does not change for $\Delta E_{\text{SL}} = 50 \text{ cm}^{-1}$ and 200 cm^{-1} , where we obtain nearly identical results. The values for $P = 100 \text{ bar}$ are virtually independent of the water concentration, because the collisional deactivation of the intermediate causes the decomposition to be much slower than the consecutive reaction with water. The yield ϕ_4 is still pressure-dependent (due to the pressure dependence of k^{ca}) but only weakly depends on $[\text{H}_2\text{O}]$, because $k^{\text{ca}} < k_4[\text{H}_2\text{O}]$ (transition range, see Fig. 6). The much lower yields ϕ_4 at $T = 230 \text{ K}$ are essentially due to the much lower H_2O concentration for a given relative humidity, which overcompensates the longer lifetime (see Table 3).

To get an impression on the parameter sensitivity of our results, we calculated relative yields ϕ_4 for varying angular momentum quantum numbers J and varying SACM interpolation parameters α/β . The width of the thermal rotational distribution of HOSO_4 at $T = 298 \text{ K}$ can be characterized by $J = \langle J \rangle \pm 25 = 49 \pm 25$. We obtain *e.g.*, for a relative humidity of 50% and a pressure of 1 bar the following values for ϕ_4 : 1.6×10^{-3} ($J = 24$), 1.5×10^{-3} ($J = 49$, *cf.* Table 4), and 1.1×10^{-3} ($J = 74$). Similarly small variations are observed for the other conditions in Table 4 showing the adequacy of our approach to use specific rate coefficients with averaged J . The influence of variations in α/β on ϕ_4 is almost exclusively restricted to reaction (2b) and much less to reaction (4). Here it is important to note that an increase of α/β for reaction (2b) increases the rate of reaction (2b) and, in this way, further lowers the yield ϕ_4 . So the cases of interest for atmospheric chemistry are the cases where α/β for reaction (2b) is lowered and the yield ϕ_4 is increased. Here, we obtain, again for a relative humidity of 50% and a pressure of 1 bar, the following values for ϕ_4 : 1.5×10^{-3} ($\alpha/\beta = 0.5$, *cf.* Table 4), 6.1×10^{-3} ($\alpha/\beta = 0.4$), 3.3×10^{-2} ($\alpha/\beta = 0.3$), and 1.8×10^{-1} ($\alpha/\beta = 0.2$). It is obvious that only for values of $\alpha/\beta < 0.3$ the yield ϕ_4 is significantly increased. However, such small values, though not generally impossible, are unlikely for reaction (2b) as was outlined in section 2.2. Here, more work on the $\text{HO}_2 + \text{SO}_3$ reaction is clearly needed. We note that $\alpha/\beta = 0.2$ corresponds to a high-pressure limiting rate coefficient $k_{-2b}^{\infty}(T = 298 \text{ K}) = 4.9 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$, at least three orders of magnitude below the collision limit.

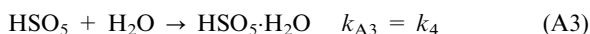
We finally note once more that our calculated yields of reaction (4) are likely to represent upper limits, because the high-pressure limiting value for k_4 was used. If this approximation is dropped, the yields will even be smaller, and an additional pressure dependence may arise from a pressure dependence of k_4 . Summarizing, we conclude that reaction (4) with a relative yield below 1% is very unlikely to represent a significant pathway to atmospheric aerosols.

4. Conclusions

We performed a theoretical study on the atmospheric fate of the HSO_5 radical produced in the $\text{HOSO}_2 + \text{O}_2$ reaction. Special attention was paid to its consecutive reaction with H_2O as a possible route to atmospheric aerosols. To this end a chemical activation master equation with a bimolecular sink term was set up and solved. Different steady-state regimes were examined as well as the relation between chemically and

thermally activated decomposition of HSO_5 . We obtained an upper limit of 1% for the relative yield of the $\text{HSO}_5 + \text{H}_2\text{O}$ reaction under tropospheric conditions, which rules out this reaction as a significant pathway to atmospheric aerosols.

Our results can be approximately incorporated in atmospheric-chemical models by using the following mechanism and rate coefficients:



with ϕ_4 , k_2 , and k_4 as given above. This simplified steady-state model is valid as long as $\phi_4 k_2 [\text{O}_2] \ll k_4 [\text{H}_2\text{O}]$.

Acknowledgements

N.G.-G. is grateful to the Alexander von Humboldt Foundation for a Research Fellowship. M. O. thanks Dr. T. Berndt for stimulating discussions.

References

- J. H. Seinfeld and S. N. Pandis, *Atmospheric Chemistry and Physics*, Wiley, New York, 1998.
- B. J. Finlayson-Pitts and J. N. Pitts, Jr., *Chemistry of the Upper and Lower Atmosphere*, Academic Press, San Diego, 2000.
- M. Kulmala, *Science*, 2003, **302**, 1000.
- D. D. Davis, A. R. Ravishankara and S. Fischer, *Geophys. Res. Lett.*, 1979, **6**, 113.
- W. R. Stockwell and J. G. Calvert, *Atmos. Environ.*, 1983, **17**, 2231.
- D. Majumdar, G.-S. Kim, J. Kim, K. S. Oh, J. Y. Lee, K. S. Kim, W. Y. Choi, S.-H. Lee, M.-H. Kang and B. J. Min, *J. Chem. Phys.*, 2000, **112**, 723.
- N. González-García, W. Klopfer and M. Olzmann, *Chem. Phys. Lett.*, 2009, **470**, 59.
- T. Berndt, O. Böge, F. Stratmann, J. Heintzenberg and M. Kulmala, *Science*, 2005, **307**, 698.
- T. Berndt, F. Stratmann, S. Bräsel, J. Heintzenberg, A. Laaksonen and M. Kulmala, *Atmos. Chem. Phys.*, 2008, **8**, 6365.
- A. Laaksonen, M. Kulmala, T. Berndt, F. Stratmann, S. Mikkonen, A. Ruuskanen, K. E. J. Lehtinen, M. Dal Maso, P. Aalto, T. Petäjä, I. Riipinen, S.-L. Sihto, R. Janson, F. Arnold, M. Hanke, J. Ücker, B. Umann, K. Sellegri, C. D. O'Dowd and Y. Viisanen, *Atmos. Chem. Phys.*, 2008, **8**, 7255.
- H. Du and F. Yu, *Atmos. Chem. Phys.*, 2009, **9**, 7913.
- J. Gonzalez, M. Torrent-Sucarrat and J. M. Anglada, *Phys. Chem. Chem. Phys.*, 2010, **12**, 2116.
- J. P. Friend, R. A. Barnes and R. M. Vasta, *J. Phys. Chem.*, 1980, **84**, 2423.
- T. Kurtén, T. Berndt and F. Stratmann, *Atmos. Chem. Phys.*, 2009, **9**, 3357.
- M. Sipilä, T. Berndt, T. Petäjä, D. Brus, J. Vanhanen, F. Stratmann, J. Patokoski, R. L. Mauldin III, A.-P. Hyvärinen, H. Lihavainen and M. Kulmala, *Science*, 2010, **327**, 1243.
- J. J. Margitan, *J. Phys. Chem.*, 1984, **88**, 3314.
- D. Martin, J. L. Jourdain and G. Le Bras, *J. Phys. Chem.*, 1986, **90**, 4143.
- J. F. Gleason, A. Sinha and C. J. Howard, *J. Phys. Chem.*, 1987, **91**, 719.
- J. F. Gleason and C. J. Howard, *J. Phys. Chem.*, 1988, **92**, 3414.
- W. Klopfer, D. P. Tew, N. González-García and M. Olzmann, *J. Chem. Phys.*, 2008, **129**, 114308.
- R. G. Parr and W. Yang, *Density-functional theory of atoms and molecules*, Oxford University Press, Oxford, 1989.
- C. Lee, W. Yang and R. G. Parr, *Phys. Rev. B: Condens. Matter*, 1988, **37**, 785.
- A. D. Becke, *Chem. Phys.*, 1993, **98**, 5648.
- T. H. Dunning, *J. Chem. Phys.*, 1989, **90**, 1007.
- J. P. Merrick, D. Moran and L. Radom, *J. Phys. Chem. A*, 2007, **111**, 11683.
- J. Čížek, *J. Chem. Phys.*, 1966, **45**, 4256.
- J. D. Purvis III and R. J. Bartlett, *J. Chem. Phys.*, 1982, **76**, 1910.
- J. A. Pople, M. Head-Gordon and K. Raghavachari, *J. Chem. Phys.*, 1987, **87**, 5968.
- T. Helgaker, W. Klopper, H. Koch and J. Noga, *J. Chem. Phys.*, 1997, **106**, 9639.
- A. Halkier, T. Helgaker, P. Jorgensen, W. Klopper, H. Koch, J. Olsen and A. K. Wilson, *Chem. Phys. Lett.*, 1998, **286**, 243.
- M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb and J. R. Cheeseman, J. A. Montgomery, Jr., T. Vreven, K. N. Kudin, J. C. Burant, J. M. Millam, S. S. Iyengar, J. Tomasi, V. Barone, B. Mennucci, M. Cossi, G. Scalmani, N. Rega, G. A. Petersson, H. Nakatsuji, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, M. Klene, X. Li, J. E. Knox, H. P. Hratchian, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, P. Y. Ayala, K. Morokuma, G. A. Voth, P. Salvador, J. J. Dannenberg, V. G. Zakrzewski, S. Dapprich, A. D. Daniels, M. C. Strain, O. Farkas, D. K. Malick, A. D. Rabuck, K. Raghavachari, J. B. Foresman, J. V. Ortiz, Q. Cui, A. G. Baboul, S. Clifford, J. Cioslowski, B. B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R. L. Martin, D. J. Fox, T. Keith, M. A. Al-Laham, C. Y. Peng, A. Nanayakkara, M. Challacombe, P. M. W. Gill, B. Johnson, W. Chen, M. W. Wong, C. Gonzalez and J. A. Pople, *Gaussian 03, Revision C.02*, Gaussian, Inc., Wallingford, CT, 2004.
- E. T. Aaltonen and J. S. Francisco, *J. Phys. Chem. A*, 2003, **107**, 1216.
- R. G. Gilbert and S. C. Smith, *Theory of Unimolecular and Recombination Reactions*, Blackwell, Oxford, 1990.
- K. A. Holbrook, M. J. Pilling and S. H. Robertson, *Unimolecular Reactions*, Wiley, Chichester, 2nd edn, 1996.
- M. Olzmann, J. Gebhardt and K. Scherzer, *Int. J. Chem. Kinet.*, 1991, **23**, 825.
- M. Olzmann, *Phys. Chem. Chem. Phys.*, 2002, **4**, 3614.
- A. Maranzana, J. R. Barker and G. Tonachini, *Phys. Chem. Chem. Phys.*, 2007, **9**, 4129.
- F. Caralp, W. Forst and A. Bergeat, *Phys. Chem. Chem. Phys.*, 2008, **10**, 5746.
- W. H. Press, B. P. Flannery, S. A. Teukolsky and W. T. Vetterling, *Numerical Recipes in Fortran*, Cambridge University Press, Cambridge, 2nd edn, 1992.
- B. T. Smith, J. M. Boyle, B. S. Garbow, Y. Ikebe, V. C. Klema and C. B. Moler, *Matrix Eigensystem Routines—EISPACK Guide*, ed. G. Goos and J. Hartmanis, Lecture Notes in Computer Science, Springer, Berlin, 1974, vol. 6.
- J. Troe, *J. Chem. Phys.*, 1981, **75**, 226.
- J. Troe, *J. Chem. Phys.*, 1983, **79**, 6017.
- T. Baer and W. L. Hase, *Unimolecular Reaction Dynamics*, Oxford University Press, New York, 1996.
- C. J. Cobos and J. Troe, *J. Chem. Phys.*, 1985, **83**, 1010.
- J. Troe, *Z. Phys. Chem. Neue Folge*, 1989, **161**, 209.
- M. Solimannejad, G. Azimi and L. Pejov, *Chem. Phys. Lett.*, 2004, **391**, 201.
- J. Troe, *J. Chem. Soc., Faraday Trans.*, 1994, **90**, 2303.
- T. Beyer and D. F. Swinehart, *Commun. ACM*, 1973, **16**, 379.
- D. C. Astholz, J. Troe and W. Wieters, *J. Chem. Phys.*, 1979, **70**, 5107.
- B. S. Rabinovitch and R. W. Diesen, *J. Chem. Phys.*, 1959, **30**, 735.
- L. I. Stiel and G. Thodos, *AIChE J.*, 1964, **10**, 266.
- J. E. Stuckey and C. H. Secoy, *J. Chem. Eng. Data*, 1963, **8**, 386.
- R. C. Reid, J. M. Prausnitz and B. E. Poling, *The Properties of Gases and Liquids*, McGraw-Hill, Boston, 4th edn, 1987.
- N. Snider, *J. Phys. Chem.*, 1986, **90**, 4366.
- J. Troe, *J. Phys. Chem.*, 1983, **87**, 1800.
- J. Troe, *Z. Phys. Chem. Neue Folge*, 1987, **154**, 73.
- J. Troe, *J. Chem. Phys.*, 1977, **66**, 4745.
- I. Oref and D. C. Tardy, *Chem. Rev.*, 1990, **90**, 1407.
- H. W. Schranz and S. Nordholm, *Chem. Phys.*, 1984, **87**, 163.
- N. Snider, *J. Chem. Phys.*, 1984, **80**, 1885.
- H. O. Pritchard and S. R. Vatsya, *J. Phys. Chem.*, 1984, **88**, 5816.
- S. C. Smith, M. J. McEwan and R. G. Gilbert, *J. Chem. Phys.*, 1989, **90**, 4265.
- CRC Handbook of Chemistry and Physics*, ed. D. R. Lide, CRC Press, Boca Raton, FL, 75th edn, 1994.